Lecture 12 (10/12)

Study design, confounding, and multiple predictors

Unit 3 summary:

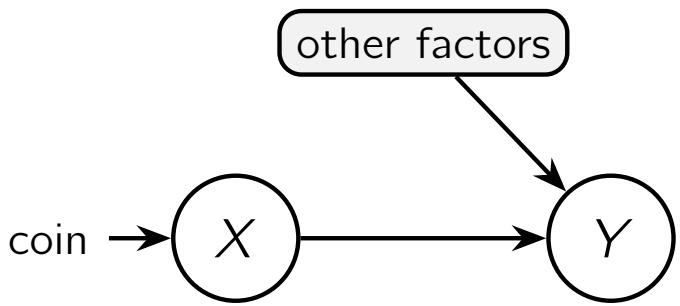
- An estimator $\hat{\theta}$ is how we approximate a parameter θ from data. Its variation over replicates of the data (repeated experiments, different samples) is the sample distribution.
- The error breaks down into bias and variance, $\text{MSE}_{\hat{\theta}} = \text{se}(\hat{\theta})^2 + \text{Bias}^2$. We look for consistent estimators where both vanish.
- Confidence intervals and p-values are computed from the sample distribution.
- The least squares estimators $\hat{\beta}_0, \hat{\beta}_1$ provide unbiased estimates of parameters in a regression model (assuming data points are sampled independently). $R^2 = \hat{\rho}^2$ is the estimate of ρ^2.
- Autoregressive models are a type of regression model where the next value in the time series is the response variable. Here, the (X, Y) points are successive values (Y_t, Y_{t+1}) and the pairs are therefore not independent, leading to a situation where least squares estimators are biased.

Two ways to collect data

Does a treatment ($X = 1$) change an outcome Y, compared with no treatment ($X = 0$)?

Randomized controlled trial (RCT)

we assign X by a coin flip



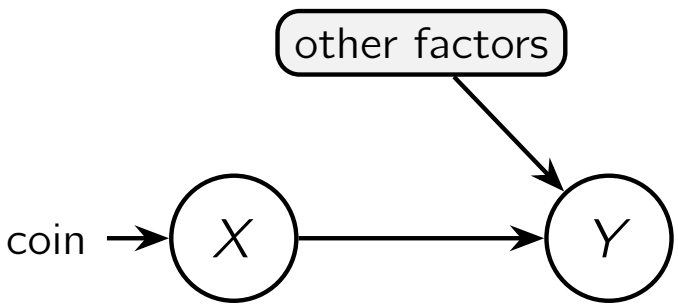
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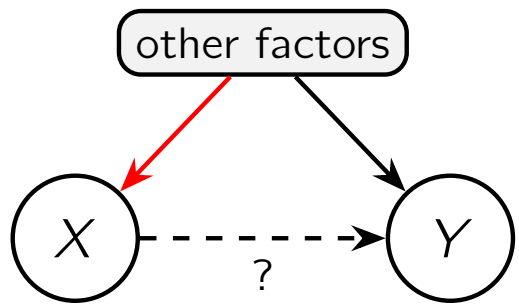
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Cohort study (observational)

we follow people and observe who gets X

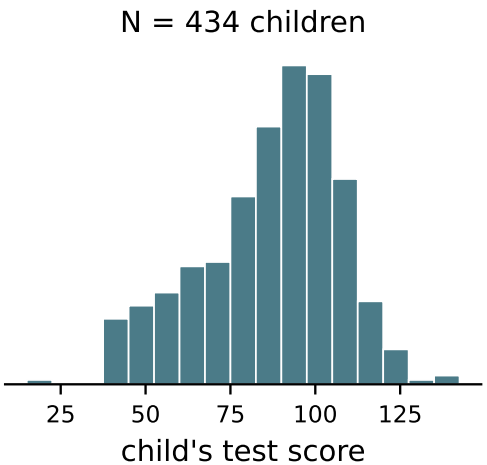


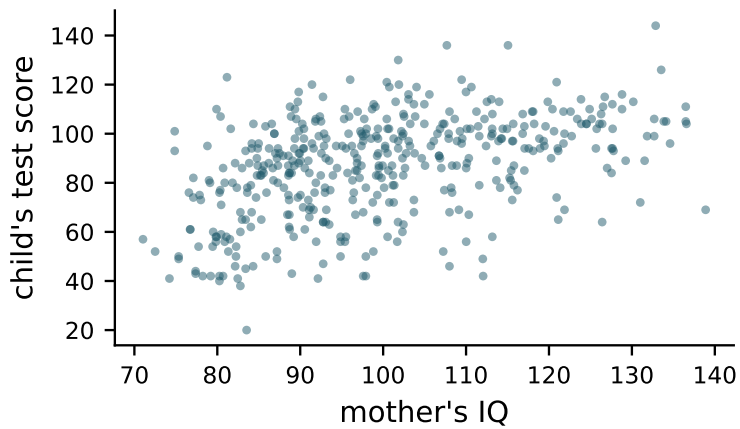
X may be correlated with other factors that also affect Y, so we cannot tell what causes what.

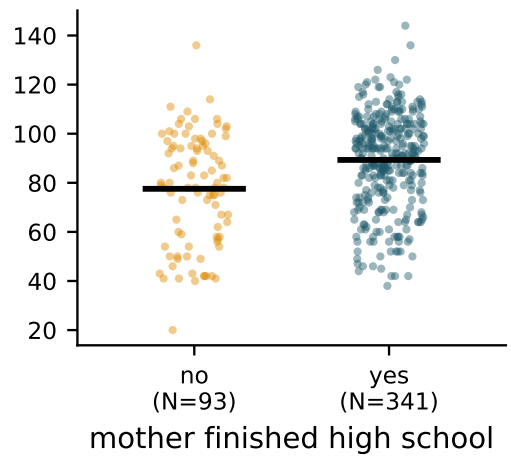
Example: children's test scores

Test scores of $N = 434$ children, with two facts about each child's mother:

- Y: the child's test score;
- $X_{\text{hs}} = 1$ if the mother finished high school, 0 if not;
- X_{iq}: the mother's IQ.



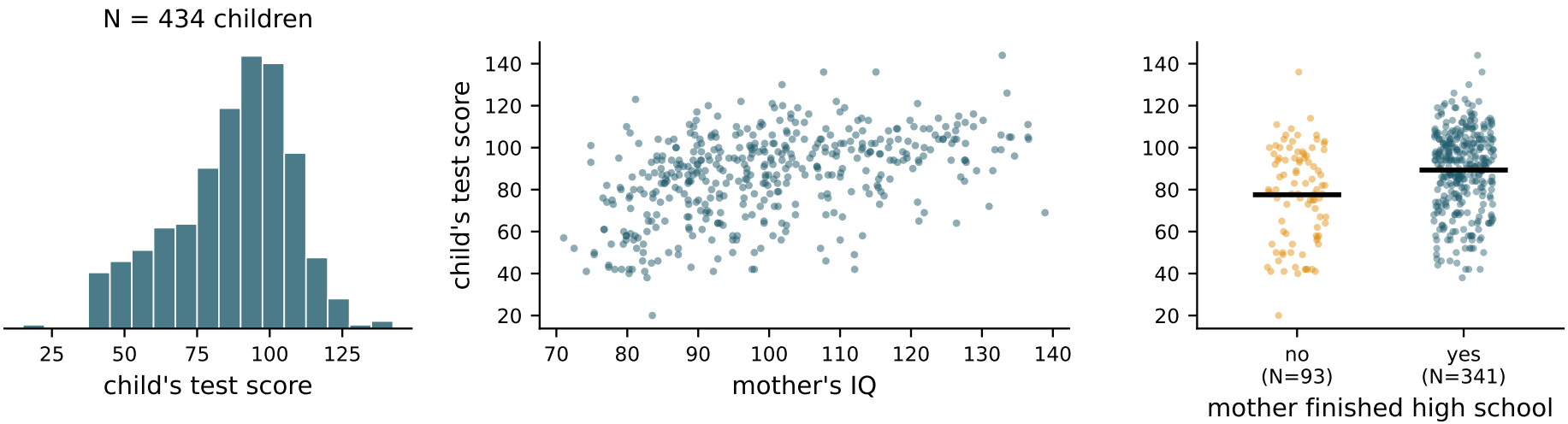




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Mothers who finished high school also tend to have higher IQs (average 102 vs. 92), so comparing the two groups alone mixes the two effects.

A model with two predictors

To compare children whose mothers have the *same* IQ, put both variables in the model:

$$Y = \beta_0 + \beta_{\text{hs}} X_{\text{hs}} + \beta_{\text{iq}} X_{\text{iq}} + \epsilon, \qquad \epsilon \sim \text{Normal}(0, \sigma_\epsilon^2).$$

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We say β_{hs} is the effect of schooling controlling for IQ (and vice versa).

Fitting the model in Python: `statsmodels`

The `statsmodels` package fits regression models and reports the estimates and their standard errors:

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import statsmodels.api as sm

X = sm.add_constant(np.column_stack([mom_hs, mom_iq]))
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Dep. Variable:                      y   R-squared:                       0.214
No. Observations:                 434   Adj. R-squared:                  0.210
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
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mom_hs         5.9501      2.212      2.690      0.007       1.603      10.297
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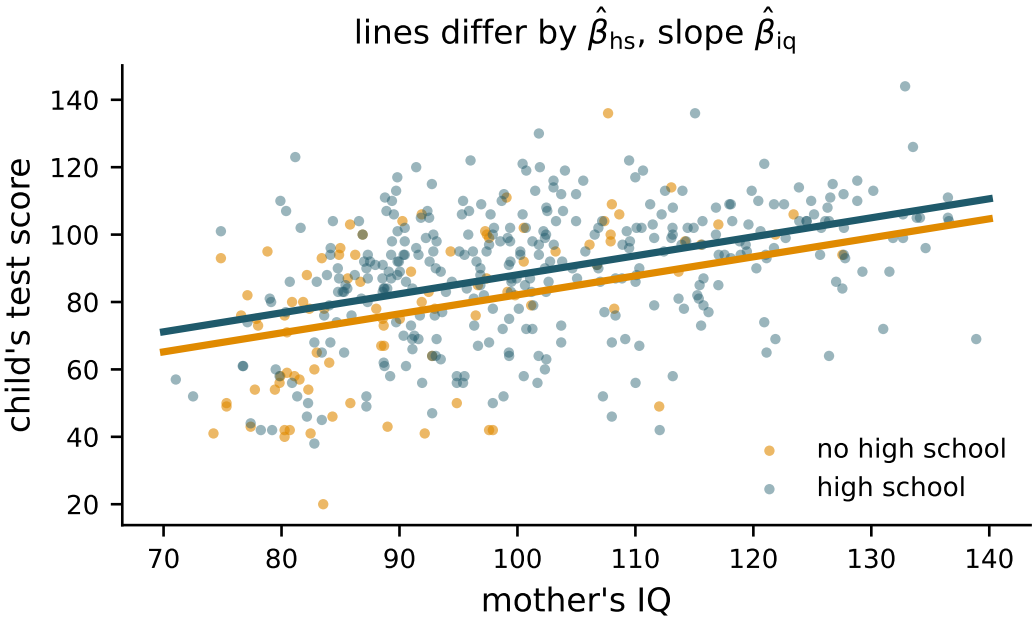
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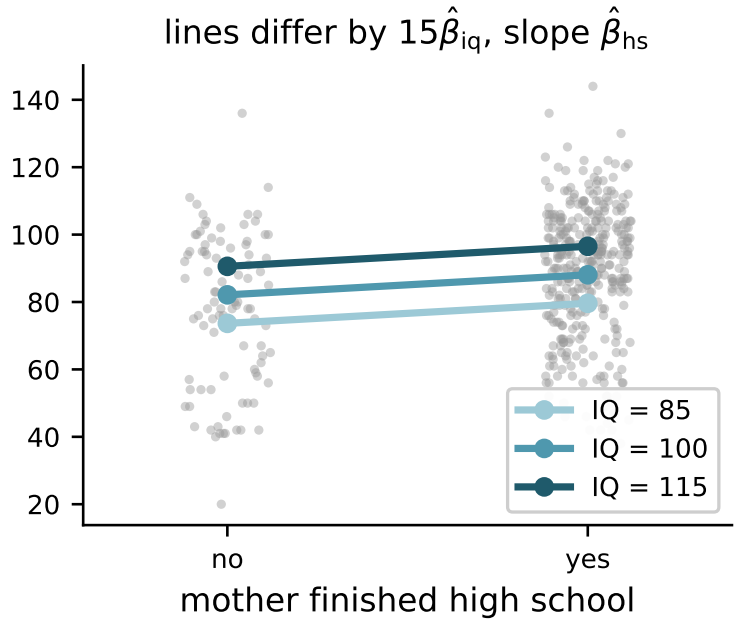
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`coef`: $\hat{\beta}$; `std err`: se($\hat{\beta}$); `t`: $\hat{T} = \hat{\beta}/\text{se}$; `P>|t|`: p-value; last two columns: 95%-CI. (`add_constant` adds the intercept β_0.)

Visualizing the fit

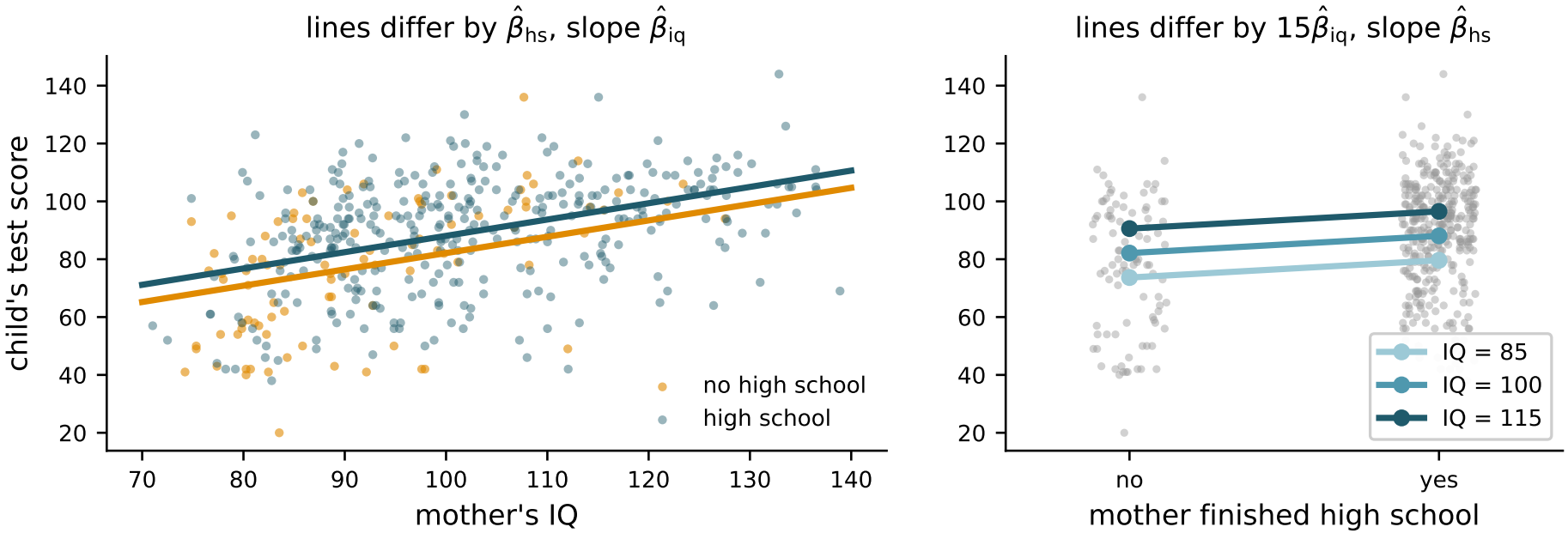
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For mothers with the *same* IQ, high school goes with $\hat{\beta}_{\text{hs}} \approx 6$ more points; for mothers with the *same* schooling, each IQ point goes with $\hat{\beta}_{\text{iq}} \approx 0.56$ more points. $R^2 = 0.21$: together they explain about 21% of the variation in scores.

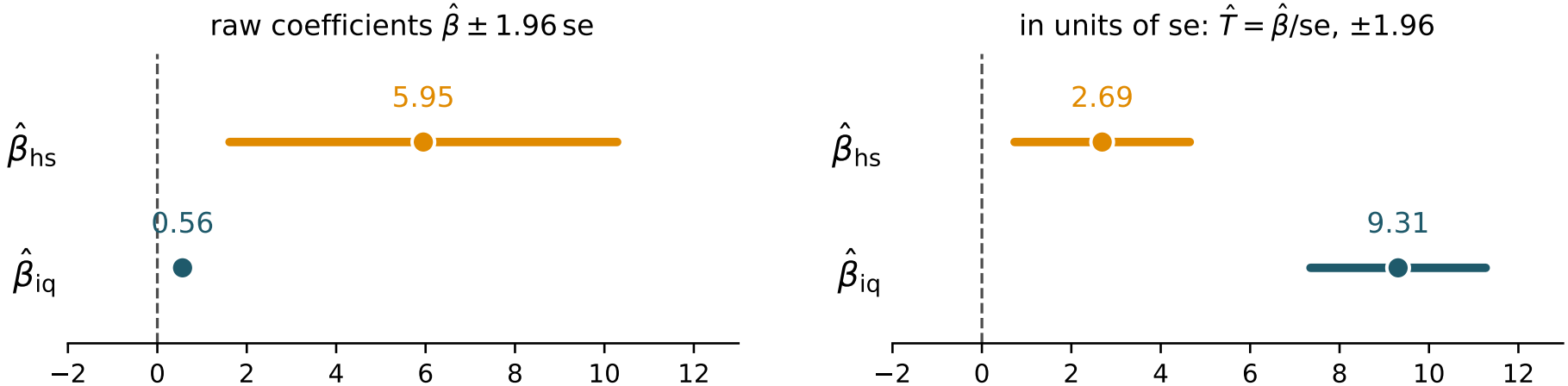
Which predictor matters more?

$\hat{\beta}_{\text{hs}} = 5.95$ is ten times $\hat{\beta}_{\text{iq}} = 0.56$, but the coefficients have different units (points for finishing high school vs. points per IQ point), so we cannot compare them directly.

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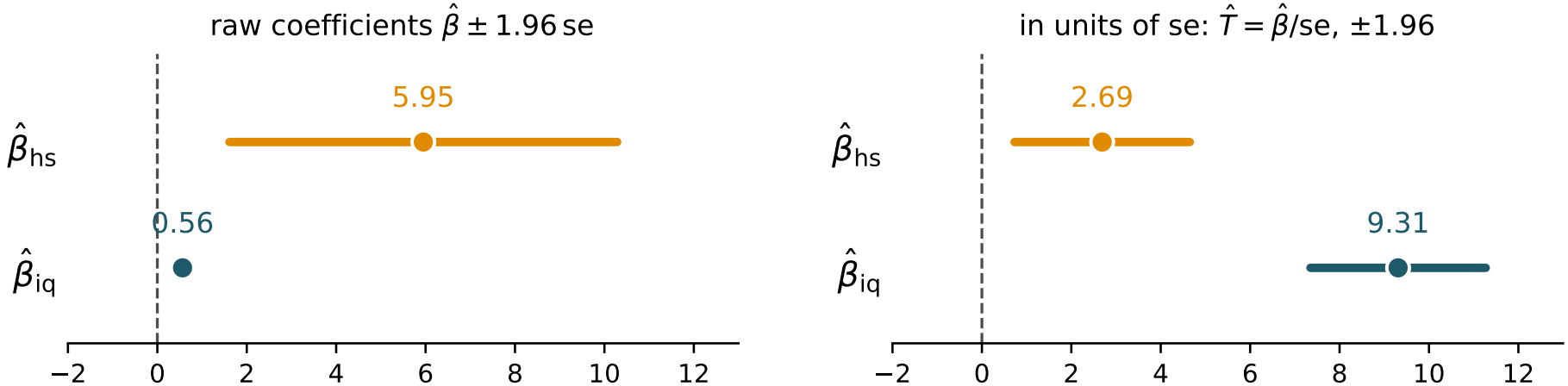
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$\hat{T}_{\text{iq}} = 9.3$ vs. $\hat{T}_{\text{hs}} = 2.7$: the IQ coefficient is far more clearly different from 0 relative to its uncertainty.

R^2 for the test scores

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predictors	R^2
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- About 79% of the variation is *not* explained: the scores have standard deviation 20, and the residuals still have $\hat{\sigma}_\epsilon \approx 18$. The model says a lot about *average* scores, but little about any one child.

Multiple-predictor regression: notation

In general, with K predictors $X_1, \ldots, X_K$ and a real-valued response Y,

$$Y = \beta_0 + \sum_{k=1}^{K} \beta_k X_k + \epsilon, \qquad \epsilon \sim \text{Normal}(0, \sigma_\epsilon^2),$$

or equivalently $Y \mid X_1, \ldots, X_K \sim \text{Normal}\left(\beta_0 + \sum_{k=1}^{K} \beta_k X_k,\ \sigma_\epsilon^2\right)$.

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The subscript on X now has a new meaning:

- X_k is the k-th *predictor* (e.g. $X_1 = X_{\text{hs}}$, $X_2 = X_{\text{iq}}$);
- $X_{k,i}$ is the value of the k-th predictor for the i-th *data point*, $i = 1, \ldots, N$, and Y_i is its response.

Interpreting the regression coefficients

β_k is the difference in average Y per unit of X_k *with all other predictors held fixed*:

$$\beta_k = E[Y \mid \ldots, X_k = x + 1, \ldots] - E[Y \mid \ldots, X_k = x, \ldots],$$

where "..." are the same values of the other predictors on both sides. We say β_k is the effect of X_k controlling for the other predictors.

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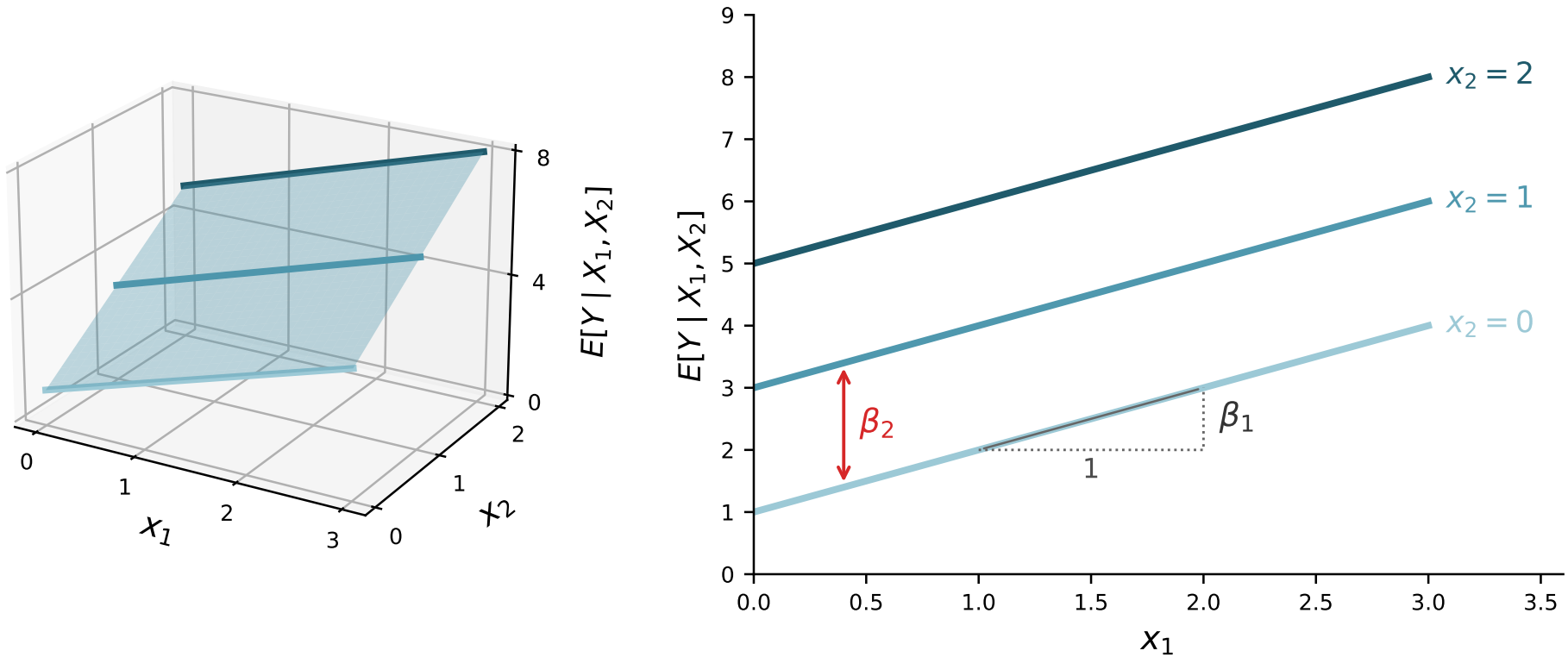
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Linearity means this difference is the same whatever values the other predictors are held at.

The regression plane

With two predictors, $E[Y \mid X_1, X_2] = \beta_0 + \beta_1 X_1 + \beta_2 X_2$ is a plane:



Fixing x_2 slices the plane into parallel lines: each has slope β_1, and they are spaced by β_2 (the left panel of the test-score figure is exactly this picture).

R^2 with multiple predictors

R^2 is defined exactly as before,

$$R^2 = 1 - \frac{\text{SSR}}{\text{SST}}, \qquad r_i = Y_i - \left(\hat{\beta}_0 + \hat{\beta}_1 X_{1,i} + \cdots + \hat{\beta}_K X_{K,i}\right),$$

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and has the same interpretation: the fraction of the variation in Y explained by the predictors, now *all* of them together,

$$R^2 \approx 1 - \frac{\text{var}(Y \mid X_1, \ldots, X_K)}{\text{var}(Y)}.$$